

a) $sVal = sVal1 / sVal$;

```
mov ax, sVal1    ; ax = sVal1
cwd              ; dx: ax
idiv sVal         ; dx = remainder, ax = quotient
mov sVal, ax      ; sVal = ax
```

b) $sVal = sVal1 \% sVal2$;

```
mov ax, sVal1    ; ax = sVal1
cwd              ; dx: ax
idiv sVal2       ; dx = rem., ax = quo.
mov sVal, dx     ; sVal = dx
```

c) $iVal = sVal1 / iVal1$;

```
mov ax, sVal1    ; ax = sVal1
cwde             ; eax
cdq              ; edx: eax
idiv iVal1       ; edx = rem., eax = quo.
mov iVal, eax    ; iVal = eax
```

d) iVal = iVal1 / bVal2;

```
mov eax, iVal1      ; eax = iVal1
idiv bVal2          ; ax/bVal2 → ah=rem., al=quo.
cbw                 ; ax = al
cwd                 ; eax = ax
mov iVal, eax        ; iVal = eax
```

e) sVal = iVal1 % bVal;

```
mov eax, iVal1      ; eax = iVal1
idiv bVal           ; ax/bVal → ah=rem., al=quo.
mov al, ah          ; al = ah
cbw                 ; ax = al
mov sVal, ax        ; sVal = ax
```

f) iVal = sVal1 % iVal2;

```
mov ax, sVal1       ; ax = sVal1
cwd                 ; eax = ax
cdq                 ; edx:eax = eax
idiv iVal2           ; edx=rem., eax=quo.
mov iVal, edx        ; iVal = edx
```

g) iVal = bVal1 / bVal2;

```
mov al, bVal1       ; al = bVal1
cbw                 ; ax = al
idiv bVal2          ; ah=rem., al=quo.
cbw                 ; ax = al
cwd                 ; eax = ax
mov iVal, eax        ; iVal = eax
```